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March 26, 2026

Load Balancer with NAT Gateway Implementation in Microsoft Azure

Screenshots

Resource group NAT-RG created in East US to house all lab resources.

☐	Name	Type	Location	Resource Group	Subscription	Last accessed
☐	NAT-RG	Resource group	East US	NAT-RG	Azure subscription 1	31 seconds ago

VNet-NAT set up with address space 10.0.0.0/16 and one subnet for both VMs.

Home

VNet-NAT

Virtual network

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Move

Delete

Refresh

Give feedback

Essentials

Resource group (move) : NAT-RG

Location (move) : East US

Subscription (move) : Azure subscription 1

Subscription ID : [redacted]

Tags (edit) : Add tags

Address space : 10.0.0.0/16

Subnets : 1 subnet

DNS servers : Azure provided DNS service

BGP community string : Configure

Virtual network ID : [redacted]

Troubleshoot

Properties

Capabilities (5)

Recommendations

Tutorials

Both Ubuntu-Web1 and Ubuntu-Web2 deployed and running inside NAT-RG with no public IP.

Ubuntu-Web1

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Help me copy this VM in any region

Manage this VM with Azure CLI

Connect

Start

Restart

Stop

Hibernate

Capture

Delete

Refresh

Scale

Open in mobile

Feedback

CLI

Essentials

Resource group (move) : NAT-RG

Status : Running

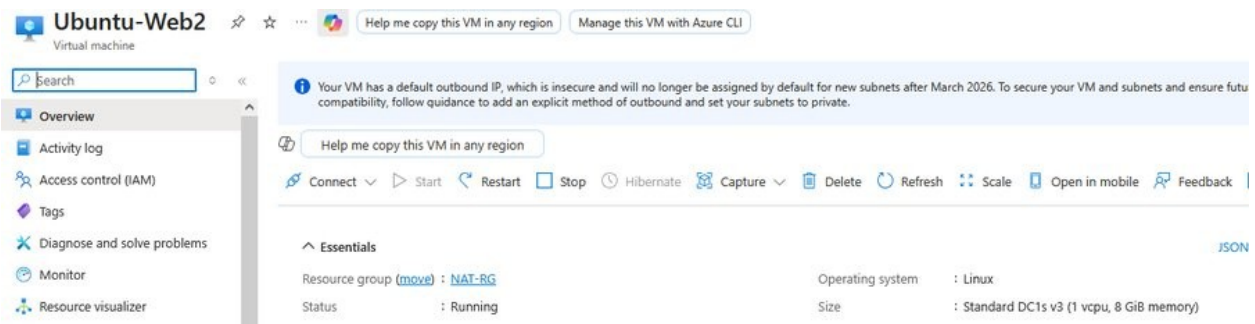
Location : East US

Operating system : Linux (ubuntu 24.04)

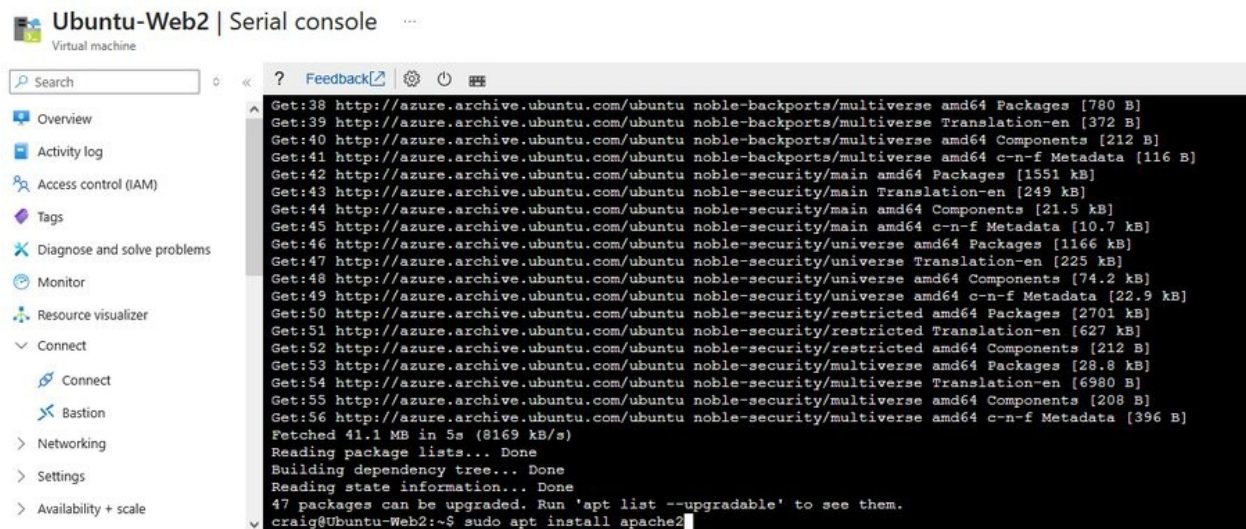
Size : Standard DC1s v3 (1 vcpu, 8 GiB memory)

Primary NIC public IP : -

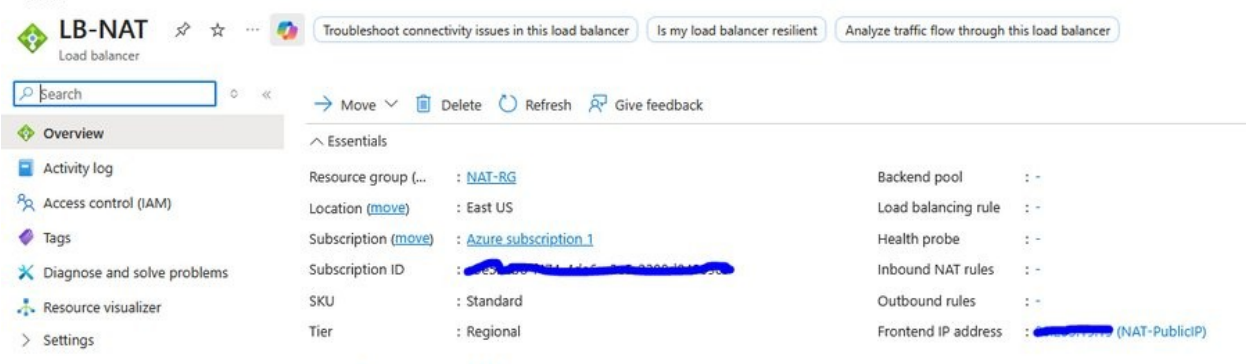
Ubuntu-Web1 overview showing Running status, East US region, and Standard DC1s v3 size.



LB-NAT load balancer created with Standard SKU and a static frontend public IP.



Apache installed on Ubuntu-Web2 through the Azure Serial Console.



Ubuntu-Web2 overview showing Running status inside NAT-RG.

LB-NAT | Backend pools

Load balancer

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Frontend IP configuration

Backend pools

Health probes

+ Add

Refresh

Export to CSV

The backend pool is a critical component of the load balancer. The backend pool defines the group of resources that will serve traffic for a given load-balancing rule. [Learn more.](#)

Search

Add filter

Backend pool	Resource Name	IP address	Network interface	Availability zone	Rules count	Resource Status	Admin state
Backend-Pool (2)							
Backend-Pool	Ubuntu-Web1	10.0.0.4	ubuntu-web1976	-	0	Running	None
Backend-Pool	Ubuntu-Web2	10.0.0.5	ubuntu-web2797_z1	1	0	Running	None

Ubuntu-Web1 overview showing Running status with no public IP assigned.

LB-NAT | Inbound NAT rules

Name	Frontend IP	Frontend port/range	Target	Service
NAT-Web1	20.253.19.19	8080	Ubuntu-Web1	HTTP (TCP/80)
NAT-Web2	20.253.19.19	8090	Ubuntu-Web2	HTTP (TCP/80)

VNet-NAT overview confirming the virtual network and subnet configuration.

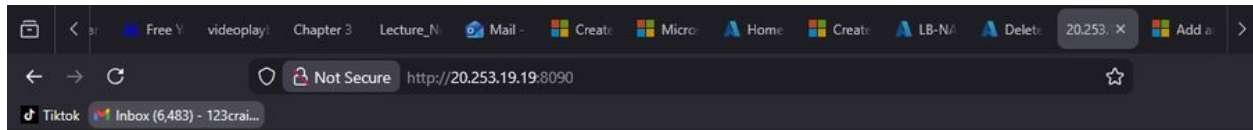
Ubuntu-Web2 | Network settings

Search rules

Source == all Destination == all Protocol == all Action == all Port == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-Web-Traffic	80,8080,8...	TCP	Any	Any	Allow

Both NAT rules configured: NAT-Web1 forwarding port 8080 and NAT-Web2 forwarding port 8090, both targeting port 80 on each VM. NSG updated to allow inbound traffic on ports 80, 8080, and 8090. Browser confirms Web server 2 is reachable at port 8090.



Web server 2



Challenges and Solutions

The biggest issue was that my school Azure subscription was disabled and marked read only. I could not create any resources at all. To fix it I created a personal Microsoft account and activated a new Azure subscription under a separate directory which got everything working.

Standard B1s did not show up in the VM size selector for East US. After some trial and error I ended up using Standard DC1s v3 which is the same size I used in a previous assignment and worked fine for this lab.

Since the VMs had no public IP I could not SSH into them directly from my machine. I used Azure Serial Console instead to run the Apache install commands which worked as a solid workaround.

What I Learned

This lab gave me a good understanding of how Azure Load Balancers work with NAT rules. The key idea is that you only need one public IP on the load balancer and then use port forwarding to direct traffic to different backend VMs. Port 8080 hits Web1 and port 8090 hits Web2 all through the same IP address.

I also learned that both the NSG rules and NAT rules have to be correctly configured. If either one is missing the traffic just times out. The backend pool also needs to be on the same VNet as the VMs otherwise the pool will not link properly.

Dealing with the disabled subscription was frustrating but it taught me how Azure directories and tenants work and how subscriptions are tied to specific directories.

Conclusion

The lab was successfully completed. Both web servers were reachable through the load balancer public IP using their respective port numbers. The setup demonstrates a practical approach to exposing multiple private backend servers through a single public endpoint using NAT rules which is a pattern commonly used in real cloud environments to minimize the number of public IPs while still being able to reach individual servers.